Department of Information Technology
Web Programming Lab Mini Project
Semester IV : Year 2025-26

Title of the Project :

Group Members

Group No	Name	Email	Phone No.	Roll No

Abstract

Keywords

Introduction

Methodology (System Flow Diagram / Model Design)

Algorithm Details (If any)

Results and Discussion (Screenshots / Charts /Figures)

Conclusion

References

Other Details if Any

Prof. Christopher Uzhuthuval
Faculty (Web Programming Lab)

Abstract –

The Hackathon Discovery and Registration Web Application is a web-based platform designed to streamline the process of organizing, discovering, and participating in hackathons. In recent years, hackathons have emerged as an important medium for innovation, collaboration, and skill development among students and professionals. However, information about such events is often scattered across multiple sources, making it difficult for interested participants to find relevant opportunities. This project aims to address this challenge by providing a centralized and user-friendly platform where users can easily access and manage hackathon-related information.

The application is developed using a combination of HTML, CSS, JavaScript, and PHP. HTML and CSS are used to design and structure the user interface, ensuring clarity and responsiveness. JavaScript is utilized to enhance interactivity and improve the overall user experience. PHP is used as the server-side scripting language to manage backend operations such as user authentication, data processing, and database interaction. The integration of these technologies results in a dynamic and efficient web application.

One of the key features of the system is its location-based search functionality, which enables users to find hackathons in their nearby areas by simply entering their city. This feature significantly reduces the time and effort required to search for relevant events. Additionally, the platform supports event registration and data storage, ensuring that all information is organized and easily retrievable.

The proposed system not only benefits participants by providing easy access to hackathons but also assists organizers in promoting their events to a wider audience without relying on manual or fragmented communication methods. It acts as a bridge between individuals seeking opportunities and those creating them.
In conclusion, the Hackathon Discovery and Registration Web Application provides a practical and efficient solution to the problem of hackathon visibility and accessibility. By combining modern web technologies with a user-centric approach, the system enhances the overall experience of discovering and participating in hackathons, thereby encouraging innovation, collaboration, and learning within the community.

Keywords - Hackathon, Web Application, Event Management System, Location-Based Search, User Registration, Event Listing, PHP, JavaScript, HTML, CSS, Database Management, Online Platform, User Interface, Event Discovery, Participant Engagement

Introduction –

In the modern digital era, web applications play a significant role in connecting people, sharing information, and providing efficient solutions to real-world problems. One such area where technology can make a meaningful impact is in the organization and discovery of hackathons. Hackathons are events where individuals or teams collaborate intensively to develop innovative solutions, enhance their technical skills, and solve practical challenges within a limited period of time. These events are widely conducted by educational institutions, organizations, and communities to promote creativity, teamwork, and problem-solving abilities.

Despite their growing popularity, information about hackathons is often not easily accessible. Many events are advertised through different platforms such as social media, websites, or internal networks, leading to scattered and unorganized information. As a result, interested participants may miss valuable opportunities simply because they are unaware of events happening in their area. Similarly, event organizers may face difficulties in reaching a wider audience and managing participant registrations effectively.

To overcome these challenges, the Hackathon Discovery and Registration Web Application has been developed as a centralized platform that bridges the gap between organizers and participants. The system allows organizers to register and publish detailed information about their hackathons, including event name, date, location, and description. At the same time, users can search for hackathons based on their city, view relevant details, and choose events that match their interests and availability.

The application is designed using standard web technologies such as HTML, CSS, JavaScript, and PHP, which ensure a structured, responsive, and interactive user experience. It focuses on simplicity, accessibility, and efficiency so that users with basic technical knowledge can easily navigate and use the platform without difficulty.

The introduction of this system aims to simplify the process of discovering and managing hackathons by providing a single, reliable source of information. It not only enhances user convenience but also encourages greater participation in hackathons by making them more visible and accessible. Overall, the project demonstrates how web-based solutions can effectively address real-world challenges and improve communication and engagement within a community.

1.1 Problem Statement

In the current digital environment, information about hackathons is widely distributed across multiple platforms such as social media, college websites, and online forums. Due to this scattered nature of information, participants often face difficulties in finding relevant hackathons based on their location and interests.

There is no single centralized system that effectively connects hackathon organizers and participants while also providing location-based filtering. As a result, users may miss important opportunities due to lack of awareness or difficulty in searching events nearby.
Additionally, organizers face challenges in reaching the right audience and managing participant registrations efficiently through manual or unstructured systems.

Therefore, there is a need for a centralized web-based platform that allows users to discover hackathons based on location and enables organizers to manage events and participant requests in an organized manner.

2. Objectives

The main objectives of the Hackathon Discovery and Registration Web Application are:

2.1 Primary Objectives:

- To develop a centralized platform for hackathon discovery and registration.
- To allow users to search hackathons based on location.
- To enable organizers to create and manage hackathon events.

2.2 Secondary Objectives:

- To implement user profile management with LinkedIn integration.
- To provide a request-based participation system.
- To improve accessibility and visibility of hackathon events.

2.3 Technical Objectives:

- To build a responsive web application using HTML, CSS, JavaScript, and PHP.
- To integrate OpenStreetMap API for location-based services.
- To ensure efficient data storage and retrieval using a database system.

Methodology –

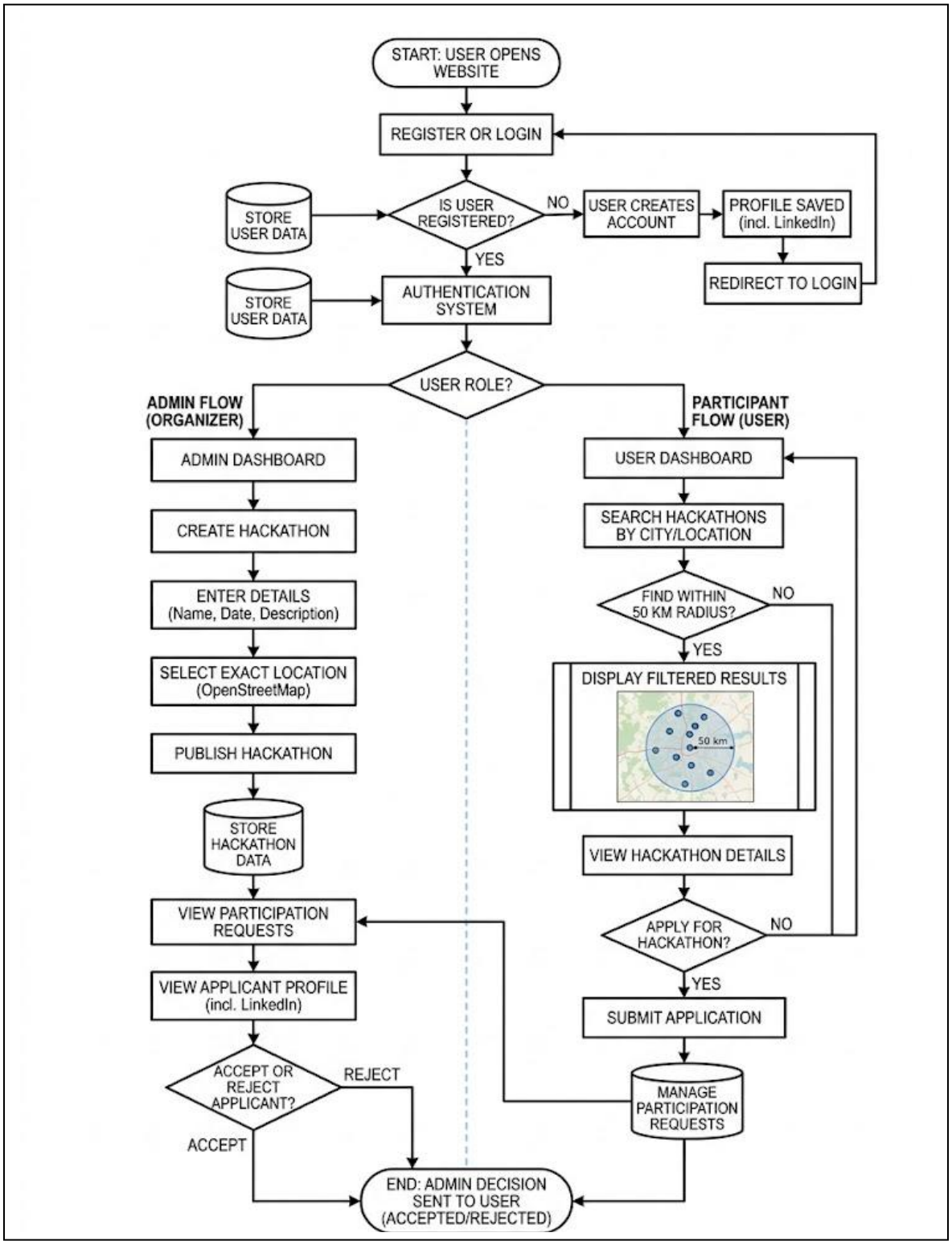
The methodology used for developing the Hackathon Discovery and Registration Web Application follows a systematic approach, including requirement analysis, design, implementation, and testing.

In the requirement analysis phase, the system is designed to support two types of users: administrators (hackathon organizers) and participants. Administrators can create and manage hackathon events, while users can search for hackathons based on their location and apply for participation. Additional features such as user profiles and LinkedIn profile linking are also considered.

During the design phase, the structure of the application is planned. The frontend is designed using HTML and CSS to provide a simple and user-friendly interface. The backend is structured using PHP to handle data processing, user authentication, and database operations.
In the implementation phase, the system is developed using HTML, CSS, JavaScript, and PHP. JavaScript is used for interactivity and form validation, while PHP manages server-side logic. The application also integrates the OpenStreetMap API to provide location-based services. Administrators can map the exact location of their hackathon, and users can search for events within a 50 km radius.

The system includes a participation request feature where users can apply for hackathons, and administrators can accept or reject requests after reviewing user profiles, including their LinkedIn links. User profile data is stored to allow quick and easy registration for future events.
Finally, the system is tested to ensure all functionalities, such as registration, login, event creation, map integration, and search features, work correctly and efficiently.
Overall, this methodology ensures a simple, functional, and user-friendly system for managing and discovering hackathons.

System Flow Diagram –



System Modules –

4. System Modules
The Hackathon Discovery and Registration Web Application is divided into several functional modules. Each module is designed to handle specific tasks, ensuring smooth operation of the system and better user experience.

4.1 User Management Module
This module handles user registration, login, and profile management. Users can create an account by providing necessary details, including their LinkedIn profile link. Once registered, users can log in and access their dashboard. This module ensures secure authentication and easy access to system features.

4.2 Admin Management Module
The admin module is responsible for managing hackathon events. Administrators can create new hackathons by entering event details such as name, description, date, and requirements. They also have the authority to update or delete events and manage participation requests submitted by users.

4.3 Hackathon Management Module
This module focuses on storing and managing hackathon-related data. It includes event creation, updating event details, and displaying hackathon information to users. All hackathon data is stored in the database for easy retrieval and management.

4.4 Location-Based Search Module
This is one of the key modules of the system. It uses OpenStreetMap API to allow users to search for hackathons based on their location. The system filters events within a 50 km radius, helping users find nearby hackathons easily and efficiently.

4.5 Participation Request Module
This module allows users to apply for hackathons they are interested in. Once a user applies, the request is sent to the admin. The admin can review the applicant's profile, including their LinkedIn information, and then decide to accept or reject the request.

4.6 Database Management Module

This module handles the storage and retrieval of all system data, including user information, hackathon details, and participation requests. It ensures data consistency, security, and efficient access during system operations.

Overall, these modules collectively ensure that the system works in an organized and efficient manner, providing a seamless experience for both organizers and participants.

System Architecture –

5. System Architecture

The Hackathon Discovery and Registration Web Application follows a client-server architecture model. The system is designed to ensure smooth communication between users (clients), the web application, and the database.

The architecture is divided into three main layers:

5.1 User Interface Layer (Frontend)

This layer is responsible for the interaction between the user and the system. It is developed using HTML, CSS, and JavaScript. Users can access features such as registration, login, searching hackathons, and applying for events through this interface. The frontend ensures that the system is simple, responsive, and user-friendly.

5.2 Application Layer (Backend)

The backend layer is developed using PHP. It handles all the business logic of the system, including user authentication, hackathon creation, processing participation requests, and location-based search functionality. This layer acts as a bridge between the frontend and the database.

5.3 Data Layer (Database)

This layer is responsible for storing and managing all system data. It includes user details, hackathon information, and participation requests. The database ensures secure and efficient storage and retrieval of data whenever required by the application.

5.4 External API Integration

The system also integrates OpenStreetMap API to provide location-based services. This API allows administrators to mark exact locations of hackathons and enables users to search for events within a 50 km radius based on their location.

5.5 System Flow Overview

1. Users access the web application through a browser.

2. Requests are sent to the backend server.
3. PHP processes the request and interacts with the database.
4. Data is retrieved and sent back to the frontend.
5. Results are displayed to the user in a structured format.

5.6 Architecture Diagram (To be Included)

A system architecture diagram should be included in this section showing:

- User (Admin / Participant)
- Frontend (HTML, CSS, JavaScript)
- Backend (PHP)
- Database
- OpenStreetMap API integration

This diagram visually represents how different components of the system interact with each other.

Overall, the system architecture ensures a well-structured flow of data and efficient communication between all components, making the application scalable, maintainable, and easy to use.

System Requirements –

6. System Requirements

The Hackathon Discovery and Registration Web Application requires both hardware and software components for proper development, deployment, and execution. These requirements ensure that the system runs efficiently and provides a smooth user experience.

6.1 Hardware Requirements

The minimum hardware requirements for running the system are as follows:

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB or above
- Storage: At least 500 MB free space for project files
- Input Devices: Keyboard and Mouse
- Display: Standard monitor with minimum resolution support
- Internet Connection: Required for accessing map services and online functionalities

These hardware requirements are sufficient to support both development and execution of the web application without performance issues.

6.2 Software Requirements

The software requirements for the development and execution of the system include:

- Operating System: Windows / Linux / macOS
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: PHP
- Database: MySQL (or any compatible relational database)
- Web Server: Apache (XAMPP / WAMP recommended for local development)
- Browser: Google Chrome / Mozilla Firefox / Microsoft Edge
- API Integration: OpenStreetMap API for location-based services

---

6.3 Functional Requirements

The functional requirements define the main features of the system:

6.3.1 User Functions

- User registration and login
- Profile creation and management
- Viewing and searching hackathons
- Applying for hackathons

6.3.2 Admin Functions

- Creating and managing hackathons
- Mapping event location using OpenStreetMap
- Viewing participant requests
- Accepting or rejecting applications

6.4 Non-Functional Requirements

The system also satisfies the following non-functional requirements:

- **Usability:** The system is simple and user-friendly for all users
- **Performance:** Fast response time for search and data retrieval
- **Security:** User authentication ensures secure access
- **Scalability:** Can support an increasing number of users and events
- **Reliability:** Ensures stable and consistent system performance

---

Overall, the system requirements ensure that the application operates efficiently and meets the expected functional and technical standards for a web-based hackathon management platform.

Results and Discussion –

The Hackathon Discovery and Registration Web Application was successfully developed and tested to ensure that all modules function as expected. The system provides a smooth and interactive experience for both administrators and participants.
The results of the system are demonstrated through various interfaces and functionalities, which are explained below. Screenshots should be included accordingly in this section to support the working of the application.

7.1 Homepage Interface
The homepage serves as the entry point of the application. It provides options for user registration and login. It is designed to be simple and user-friendly, allowing users to easily navigate to different sections of the system.
📌 *(Insert Screenshot: Homepage / Landing Page)*

7.2 User Registration and Login
The registration page allows new users to create an account by entering their details, including their LinkedIn profile link. After successful registration, users can log in to access the system.
📌 *(Insert Screenshot: Registration Page + Login Page)*

7.3 User Dashboard
After login, participants are directed to their dashboard. From here, they can search for hackathons based on location, view event details, and apply for participation.
📌 *(Insert Screenshot: User Dashboard)*

7.4 Hackathon Search and Location Feature

One of the key features of the system is location-based search. Users can find hackathons within a 50 km radius using the OpenStreetMap API. This improves accessibility and helps users discover nearby events easily.

📌 *(Insert Screenshot: Map/Search Results Page)*

7.5 Hackathon Details and Application

Users can view complete details of a selected hackathon, including description, date, and location. They can also apply for participation directly through the platform.

📌 *(Insert Screenshot: Hackathon Details Page + Apply Button)*

7.6 Admin Dashboard

The admin dashboard allows organizers to create and manage hackathons. Admins can enter event details and select exact locations using map integration.

📌 *(Insert Screenshot: Admin Dashboard + Create Hackathon Page)*

7.7 Participation Request Management

Admins can view all user requests for a hackathon. They can check user profiles, including LinkedIn information, and then accept or reject applications based on requirements.

📌 *(Insert Screenshot: Admin Request Management Page)*

7.8 Discussion

The system successfully fulfills its objective of connecting hackathon organizers and participants through a centralized platform. The integration of location-based search improves usability, while the admin control system ensures proper management of events and participants. The application performs efficiently in handling user requests, event creation, and data management.

Overall, the results demonstrate that the system is functional, reliable, and effective in achieving its intended purpose.

Conclusion-

The Hackathon Discovery and Registration Web Application was successfully developed as a centralized platform to simplify the process of finding, organizing, and participating in hackathons. The system effectively connects event organizers and participants, reducing the challenges caused by scattered information and manual registration processes.

The application provides key functionalities such as user registration, profile management with LinkedIn integration, hackathon creation by administrators, location-based search using OpenStreetMap API, and participation request management. These features ensure that users can easily discover hackathons near their location and apply for them in an efficient manner.

The use of web technologies such as HTML, CSS, JavaScript, and PHP allowed the development of a responsive and interactive system. The integration of a database ensures proper storage and retrieval of user and event data, making the system reliable and scalable.

Overall, the project demonstrates how a well-designed web application can improve accessibility, enhance user experience, and streamline event management. The system successfully meets its objectives and provides a strong foundation for future enhancements such as mobile application development and advanced recommendation features.